AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2025

Computer Science Paper II

Time: 1 hour 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offers TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 6 Marks)

i. Mention any TWO computing devices for input, output and storage. (3 Marks)

Type of Device	Device 1	Device 2
Input		
Output		
Storage		

ii. Define the following software categories: (3 Marks)

I. Freeware

_____

_____

II. Shareware

_____

_____

III. Open-Source

_____

_____

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Q.2. (Total 3 Marks)

Write any THREE differences between DRAM and SRAM.

S.No.	DRAM	SRAM
1		
2		
3		

Q.3. (Total 3 Marks)

Describe the function of each type of internal bus, i.e, data bus, address bus and control bus.

Q.4. (Total 3 Marks)

A technician is upgrading an older server to increase memory for handling more users and applications.

Based on the given scenario, answer the following questions.

i. Which memory module will be used by the technician to increase memory? (1 Mark)

ii. Give any TWO reasons in support of your stance in part i. (2 Marks)

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Q.5. (Total 3 Marks)

Define the following terms in the context of wireless communication:

i. Radio signals (1 Mark)

ii. Radio transceivers (1 Mark)

iii. Access point (1 Mark)

Q.6. (Total 3 Marks)

In a university database, there are two entities: Students and Departments.

With reference to the given information,

i. identify the type of relationship between the Student and Department entities. (1 Mark)

ii. provide an example of this relationship type. (2 Marks)

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Q.7. (Total 7 Marks)

EITHER

a.

i. Describe the function of each layer of the Transmission Control Protocol/ Internet Protocol (TCP/IP) protocol architecture. (4 Marks)

ii. Name at least ONE protocol for any three layers described in part i. (3 Marks)

OR

b.

i. Describe the term, Global Positioning System (GPS). (1 Mark)

ii. Explain the importance of GPS in any THREE points. (3 Marks)

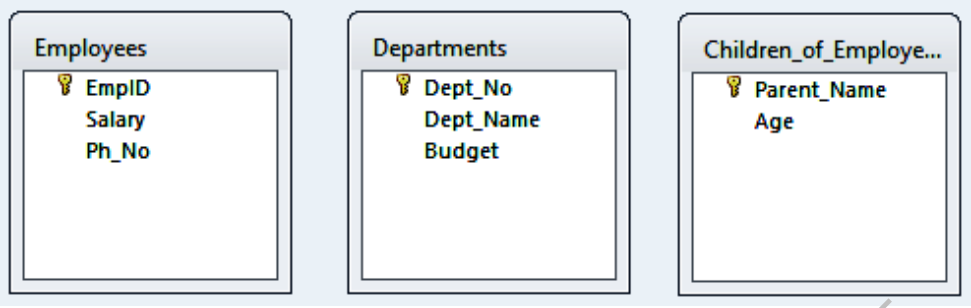
iii. Explain the THREE limitations of GPS. (3 Marks)

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Q.8. (Total 7 Marks)

EITHER

a. Consider the given image showing three tables in an MS Access database of a company.



A number of employees work in each department of the company. Each department is managed by an employee. Each child of an employee is identified by the employee's name. Furthermore, the company has the policy that only one parent can work for the company.

Draw an entity-relationship diagram (ERD) for this database. (7 Marks)

Space for ERD

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OR

b. Consider an employee management system that stores information about employees in a company. This system allows administrators to insert new employee records, retrieve employee details, and delete records.

Based on the given scenario, a table named as **Employees** has fields **Employee_ID, Name, Department, Salary, Designation, Employment_type**.

Write SQL queries for this database table to

i. insert a new record of a regular employee, Dawood_Ahmed, who joined the IT department as Associate with the salary of Rs 75000. (5 Marks)

ii. retrieve the details of all employees working in the 'Sales' department. (1 Mark)

iii. delete an employee record whose **Employee_ID is** 5. (1 Mark)

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END OF PAPER

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